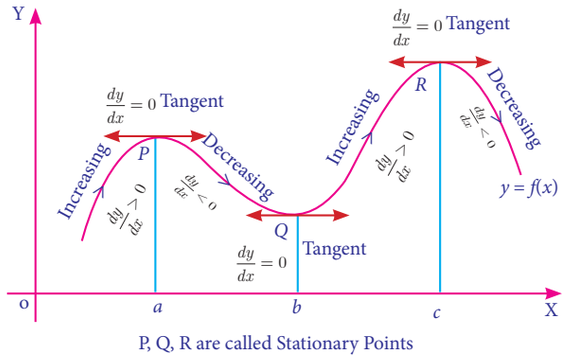
The stationary value of $f(x)$ is $f(a)$. The point $(a,f(a))$ is called stationary point.



P, Q, R are called Stationary Points

Fig: 6.9

In figure 6.9 the function $y = f(x)$ has stationary at $x = a, x = b$ and $x = c$.

At these points, $\frac{dy}{dx} = 0$. The tangents at these points are parallel to x – axis.

NOTE

By drawing the graph of any function related to economics data, we can study the trend of the business related to the function and therefore, we can predict or forecast the business trend.

Example 6.18

Show that the function $f(x) = x^3 - 3x^2 + 4x, x \in R$ is strictly increasing function on R.

Solution :

$f(x) = x^3 - 3x^2 + 4x, x \in R$

$f'(x) = 3x^2-6x+4$

$= 3x^2-6x+3+1$

$= 3(x-1)^2+1 > 0$, for all $x \in R$

Therefore, the function f is strictly increasing on $(-\infty,\infty)$.

Example 6.19

Find the interval in which the function $f(x)=x^2-4x+6$ is strictly increasing and strictly decreasing.

Solution:

Given that $f(x) = x^2-4x+6$

Differentiate with respect to x,

$f'(x) = 2x-4$

When $f'(x) = 0 \Rightarrow 2x-4=0 \Rightarrow x = 2$.

Then the real line is divided into two intervals namely $(-\infty,2)$ and $(2,\infty)$



Fig :6.10

[To choose the sign of $f'(x)$ choose any values for x from the intervals ans substitute in $f'(x)$ and get the sign.]

Interval	Sign of $f'(x) = 2x - 4$	Nature of the function
$(-\infty, 2)$	< 0	$f(x)$ is strictly decreasing in $(-\infty, 2)$
$(2, \infty)$	> 0	$f(x)$ is strictly increasing in $(2, \infty)$

Table: 6.1

Example 6.20

Find the intervals in which the function f given by $f(x)=4x^3-6x^2-72x+30$ is increasing or decreasing.

Solution :

$f(x) = 4x^3-6x^2-72x+30$

$f'(x) = 12x^2-12x-72$

$= 12(x^2-x-6)$

$= 12(x-3)(x+2)$

$f'(x) = 0 \Rightarrow 12(x-3)(x+2)=0$

$x = 3$ (or) $x = -2$

$f(x)$ has stationary at $x = 3$ and at $x = -2$.

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